Working with R

LfBD Team

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Introduction

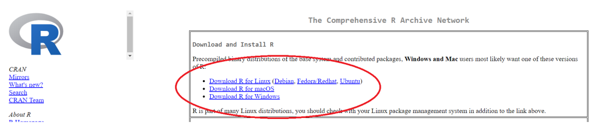
This tutorial provides the building blocks to make everyone comfortable with the environment we will be using in this course.

1. Getting started with the R programming language
2. Making your first plot in R

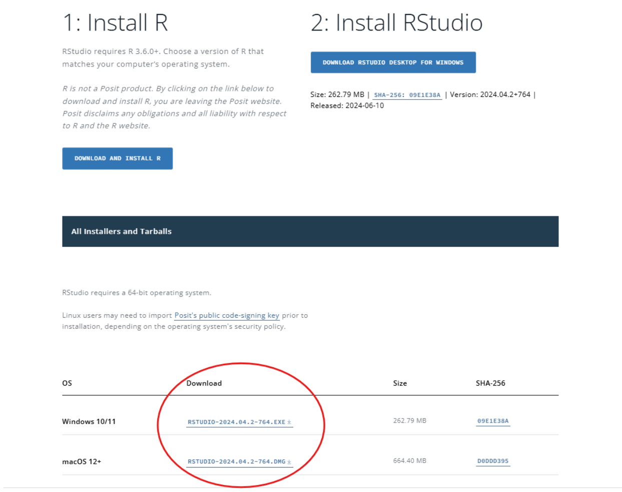
The R programming language

All of the tutorials and assignments in this course will make use of the R programming language. You might have already worked with R and have it installed. If not, go through the following steps:

1. Go to this page and select 'install R' for your operating system. The image below illustrates where you need to be.



2. After that, go to this page and download the free version of Rstudio. Again, select the installation for your operating system. The image on the next page illustrates where you should be looking.



It is outside the scope of this document to learn you everything about the R programming language. Rather, we will point to several resources one can use if (1) you want to get some experience in R before the course starts, or (2) if you are stuck with your R code.

- **Tutorials:**
  - **Syntax and data structures**: A beginner friendly introduction to R can be found here, which covers all the basic data structures and syntax. Additionally, Datacamp provides a comprehensive introductory course into R. Free tutorials can be found here.
  - **Plot making**: for an introduction to making plots in R with ggplot2, see the following overview.
  - **If you are stuck**, consult sources such as Google, stack overflow, or Youtube for troubleshooting assistance.

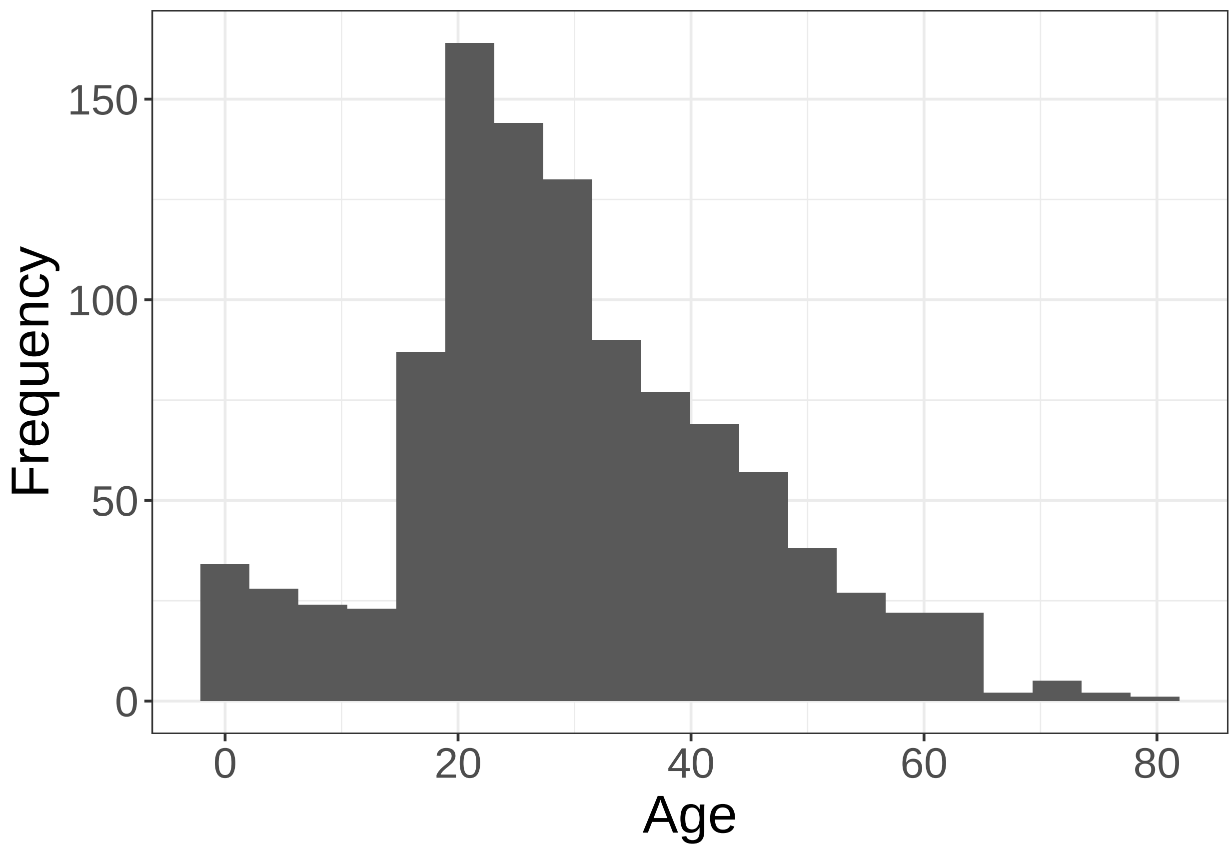
Making plots

The following is an example of how to make a plot in R. We use a dataset with information on passengers on the titanic.

```r
# Reads in the csv as an R dataframe
df_titanic <- read.csv('../input/titanic.csv')

# create ggplot object of histogram of the age of passengers
age_titanic_passengers_plot <- ggplot(data = df_titanic, # dataset to work with
                                      aes(x = Age)) + # variable for x-axis is Age
                                geom_histogram(bins=20) + # create a histogram with 20 bins
                                labs(y = 'Frequency') + # name for the Y-lab
                                theme_bw() + # theme of the plot
                                theme(text = element_text(size=20)) # increase the font size

age_titanic_passengers_plot
```


150
100
50
0
Frequency
0
20
40
60
80
Age